Gen AI names:

Claude: desktop app (paid version)

Copilot: desktop app

Prompt used:

Write a C program that chooses a fixed secret number between 1 and 10. The user should have up to 3 attempts to guess the number. Use printf and scanf, and if-else to tell the user if the guess is too high, too low, or correct. Stop immediately if the user guesses correctly and print a final win or lose message.

Claude:

```
#include <stdio.h>


int main(void) {

    int secret = 7;   // fixed secret number

    int guess;

    int attempts = 3;

    int won = 0;


    printf("Guess the number between 1 and 10. You have %d attempts.\n", attempts);


    for (int i = 1; i <= attempts; i++) {

        printf("Attempt %d/%d: ", i, attempts);

        scanf("%d", &guess);


        if (guess == secret) {

            won = 1;
```

```
            break;
        } else if (guess > secret) {
            printf("Too high!\n");
        } else {
            printf("Too low!\n");
        }
    }

    if (won) {
        printf("Correct! You guessed the number.\n");
    } else {
        printf("You lose! The number was %d.\n", secret);
    }

    return 0;
}
```

Copilot:

```
#include <stdio.h>

int main() {
    int secret = 7;      // fixed secret number between 1 and 10
    int guess;
```

```
    int attempts = 0;


    while (attempts < 3) {
        printf("Enter your guess (1-10): ");
        scanf("%d", &guess);


        if (guess == secret) {
            printf("Correct! You win!\n");
            return 0;   // stop immediately
        } else if (guess < secret) {
            printf("Too low.\n");
        } else {
            printf("Too high.\n");
        }


        attempts++;
    }


    printf("Out of attempts. You lose. The number was %d.\n", secret);
    return 0;
}
```

Correctness:
Both codes work correctly and follow the requirements. They both use the fixed number 7, give the user 3 tries, and tell the user if their guess is too high, too low, or correct. They also both stop the loop if the user guesses the number correctly.

Execution Time:
Both codes have **O(1)** execution time because the user can only guess up to 3 times. The loop will never run more than 3 times, so the amount of time the program takes stays about the same.

Space Complexity:
Both codes have **O(1)** space complexity. They only use a few integer variables and don't need more memory as the program runs.

Maintainability:
Copilot code is simpler and shorter, but Claude's more organized and makes it easier to modify later.

Program selected: Claude
Both codes are pretty simple t o read and easy to understand. I think Claude's code is a little easier to maintain because it uses the won variable to keep track of whether the user won. It also uses break to leave the loop and then handles the final win or lose message afterward.

Improvements:

I improved Claude's code by adding comments and input validation. The original code worked, but it didn't check if the user entered something that wasn't a number or entered a number outside 1 to 10. I added checks for those. Invalid inputs also don't count as one of the 3 tries. I didn't really change the main logic of the program. I mainly made it easier to use and understand.

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